NC²AN-X PPT CONTENT BIBLE

Direct Source Document for Slide Creation

Extends: NC²AN-X Ideathon Story Bible v1.0 | ISRO BAH 2026, Challenge 15

> No slide text, no bullet points, no Mermaid code appear below. This document defines what each slide must *do* — psychologically, scientifically, visually — so that whoever writes the actual words later is filling in a blueprint, not improvising one.

SECTION 1 — THE PPT STORY ARC

Beginning (Slides 1–3: Title, Team, Opportunity/USP) Job: establish who you are, that you can execute, and what's actually different — in that order. Emotional arc: *curiosity → quiet respect for the team → genuine surprise at the insight.* This is the only part of the deck where you're earning the right to be taken seriously; nothing here should ask the judge to already trust you.

Middle (Slides 4–7: Features, Process Flow, Wireframe, Architecture) Job: convert the surprise from Slide 3 into conviction that it's real and buildable. Emotional arc: *skepticism testing → reassurance → quiet confidence.* This is the engine room of the deck — the judge is now actively trying to find the hole in the idea, and every slide here should fail to give them one, without pretending the holes don't exist (see Section 9).

End (Slides 8–10: Technologies, Cost, Thank You) Job: consolidate trust and close without raising new doubts. Emotional arc: *trust consolidation → "I'd want to follow this team's progress."* Nothing exciting needs to happen here — the job of this stretch is to introduce zero new risk, not to generate a second peak.

The single emotional throughline: curiosity → respect → surprise → tested conviction → quiet confidence → trust. If any slide breaks this chain — overclaims, clutters, or contradicts an earlier one — the chain resets and the judge starts evaluating from skepticism instead of momentum.

SECTION 2 — THE REVIEWER ATTENTION MAP

Scored out of 10. Attention = how long a judge's eyes actually linger. Memorability = how likely the content survives being repeated to a co-judge later. Risk = how likely this slide actively damages the pitch if mishandled.

Slide	Attention	Memorability	Risk	What judges focus on first
1. Title	6	3	2	Whether the problem statement precisely mirrors Challenge 15's actual wording.
2. Team	4	2	3	Whether roles look real or padded.
3. Opportunity/USP	**9**	**9**	**7**	The one-liner — this is where the whole pitch is won or lost in the judge's head.
4. Features	6	5	4	Whether the feature list looks like real capabilities or a checklist.
5. Process Flow	7	6	5	Whether it's actually simple or just compressed-looking.
6. Wireframe	8	8	4	The single most visually dominant panel — almost always the heatmap.
7. Architecture	**8**	6	**8**	Whether it reads as physics-justified or jargon-stacked.
8. Technologies	3	1	6	Whether the named stack contradicts anything shown earlier.
9. Cost	5	4	3	Whether the numbers feel invented or grounded.
10. Thank You	2	1	1	Nothing — this slide is a formality.

Board reading: Slides 3 and 7 carry almost all of the deck's risk *and* almost all of its upside. That asymmetry should directly determine where design and rehearsal time go (see Section 8).

SECTION 3 — SLIDE-BY-SLIDE CONTENT STRATEGY

No slide content below — only what each slide must achieve and avoid.

Slide 1 — Title

- **Slide Purpose:** instant identity and precise problem-fit.
- **Psychological Goal:** signal seriousness before any content is read.
- **Scientific Goal:** none — pure framing.
- **Technical Goal:** none.
- **Key Message:** we answer exactly what was asked.
- **Hidden Message:** we read the brief more carefully than most teams will.
- **Visual Hierarchy:** template art → project codename → problem-statement line → leader name.
- **Information Hierarchy:** problem-statement precision first; everything else administrative.
- **Common Mistakes:** a generic title; a problem statement that paraphrases loosely instead of mirroring Challenge 15's actual wording.
- **What Must Be Removed:** any tagline that oversells before any evidence has been shown.
- **What Must Be Highlighted:** the precision of the problem-statement match.
- **What Judges Will Ask:** nothing yet.
- **Ideal Follow-Up Answer:** n/a.

Slide 2 — Team

- **Slide Purpose:** prove execution capacity.
- **Psychological Goal:** "this team can actually do this."
- **Scientific Goal:** show domain-relevant skill coverage — someone visibly owns the physics, not just the ML.
- **Technical Goal:** map each member to a real pipeline stage.
- **Key Message:** every component has an owner.
- **Hidden Message:** we divided real engineering work, not enthusiasm.
- **Visual Hierarchy:** role tag → name → college.
- **Information Hierarchy:** role-to-architecture mapping is the actual content here.
- **Common Mistakes:** undifferentiated "Member 1/2/3" entries; roles that don't map to anything shown later.
- **What Must Be Removed:** duplicate or vague role descriptions.
- **What Must Be Highlighted:** whoever owns the physics-constraint logic — this person is the credibility anchor for Slides 3 and 7.
- **What Judges Will Ask:** "who actually validated the physics?"

- **Ideal Follow-Up Answer:** name the person, name one concrete check they ran against known flare events.

Slide 3 — Opportunity / USP

- **Slide Purpose:** deliver the differentiation, honestly.
- **Psychological Goal:** the "oh, that's actually clever" reaction.
- **Scientific Goal:** communicate the Neupert → cross-attention mapping without equations.
- **Technical Goal:** state precisely what's new and what isn't.
- **Key Message:** the Story Bible's Rank-1 core-insight line.
- **Hidden Message:** we know exactly where the real novelty is, and we're not hiding behind buzzwords to inflate it.
- **Visual Hierarchy:** the one-liner → the two-path contrast graphic → supporting text.
- **Information Hierarchy:** insight first, differentiation-from-existing-ML second, USP statement last.
- **Common Mistakes:** any "first ever" / "nobody has done this" language; burying the insight under context before stating it.
- **What Must Be Removed:** every unscoped superlative.
- **What Must Be Highlighted:** the Hard=Query / Soft=Key,Value mapping, stated once, in plain language.
- **What Judges Will Ask:** "isn't attention already used for flare forecasting?"
- **Ideal Follow-Up Answer:** concede it directly — yes, broadly — then state the narrower, defensible claim: cross-modal attention bound to the Neupert relationship using Aditya-L1's own paired X-ray instruments is the specific part that's new, as far as the team could verify.

Slide 4 — Features

- **Slide Purpose:** operational concreteness.
- **Psychological Goal:** "I can picture this being used."
- **Scientific Goal:** ground each feature in a real operational need.
- **Technical Goal:** six capabilities, zero internals.
- **Key Message:** useful today, not just clever.
- **Hidden Message:** we thought past the model to the operator who has to act on it.
- **Visual Hierarchy:** icon row, deliberately equal weight across all six — this slide is an inventory, not a single insight.

- **Information Hierarchy:** ordered by operational value to a forecaster, not by build order.
- **Common Mistakes:** listing internal pipeline steps (GTI filtering, Isolation Forest) as if they were user-facing features.
- **What Must Be Removed:** anything a duty forecaster wouldn't directly care about.
- **What Must Be Highlighted:** NDI's "flags non-textbook flares" framing — the most differentiated item here.
- **What Judges Will Ask:** "what does the operator actually do when they get an alert?"
- **Ideal Follow-Up Answer:** describe the alert tiers and the action threshold that triggers escalation.

Slide 5 — Process Flow / Use-Case Diagram

- **Slide Purpose:** prove elegance and simplicity.
- **Psychological Goal:** relief — "this is simple, not a science project."
- **Scientific Goal:** trace the physical sequence (reconnection → hard X-ray → energy transfer → soft X-ray) alongside the data flow.
- **Technical Goal:** six boxes maximum, end to end.
- **Key Message:** Sun to alert, one clean line.
- **Hidden Message:** the simplicity is earned — the physics gave us this structure for free, we didn't have to invent it.
- **Visual Hierarchy:** physical-process labels → data-pipeline labels → no technical sub-detail shown at all.
- **Information Hierarchy:** physics narrative first, engineering pipeline as the scaffold beneath it.
- **Common Mistakes:** showing the full internal pipeline; presenting this as a software flowchart instead of a physics-to-alert story.
- **What Must Be Removed:** GTI, fill-value, and scaling internals — Q&A material, not slide material.
- **What Must Be Highlighted:** the single visual moment where hard X-ray "becomes" soft X-ray.
- **What Judges Will Ask:** "what happens when data is missing or delayed?"
- **Ideal Follow-Up Answer:** a brief, confident note that gap-handling and synchronization tolerance are already built into the frozen Notebook 1 — true, and worth saying plainly, without descending into filter-level detail unless pressed.

Slide 6 — Wireframes / Mockups (recommended, despite optional)

- **Slide Purpose:** make the system tangible.
- **Psychological Goal:** "I want to see this live."
- **Scientific Goal:** show the CTE heatmap actually doing its explanatory job.
- **Technical Goal:** one dominant visual, two or three supporting.
- **Key Message:** this is the screen that explains itself.
- **Hidden Message:** we designed for a human operator, not just a model output.
- **Visual Hierarchy:** CTE heatmap → flux+uncertainty panel → alert strip → everything else smaller or absent.
- **Information Hierarchy:** mirrors actual operational priority — what a duty officer would look at first during a live event.
- **Common Mistakes:** five equal-weight panels competing for attention; obviously synthetic placeholder data.
- **What Must Be Removed:** NDI gauge and multi-horizon bars as primary elements — demote both to secondary.
- **What Must Be Highlighted:** a labeled callout explaining what the heatmap's diagonal pattern physically means.
- **What Judges Will Ask:** "is this real data or a mockup?"
- **Ideal Follow-Up Answer:** be honest about build status — if it's a designed mockup, say so plainly, then point to what's already validated in Notebook 1 as the real evidence base.

Slide 7 — Architecture Diagram

- **Slide Purpose:** prove rigor without losing the room.
- **Psychological Goal:** "okay — they really thought about this."
- **Scientific Goal:** every visible block traceable to a physics or operational reason.
- **Technical Goal:** six macro-blocks, no more.
- **Key Message:** every block exists because of physics, not trend.
- **Hidden Message:** we could show sixteen more components — we're choosing not to, because we respect the judge's time.
- **Visual Hierarchy:** NC^2AN Core (cross-attention + Neupert loss) → Data Engineering → Diagnostics (CTE/NDI) → Dashboard → everything else.
- **Information Hierarchy:** physics-justification narrative first, technology labels second.
- **Common Mistakes:** the full 16-node diagram; framework-specific jargon that contradicts Slide 8's stated stack.

- **What Must Be Removed:** Isolation Forest, λ-annealing, attention-entropy regularization, GOES cross-calibration internals — Q&A only.
- **What Must Be Highlighted:** the Hard→Query, Soft→Key/Value labeling, shown explicitly on the diagram.
- **What Judges Will Ask:** "why cross-attention instead of a Transformer encoder over both streams concatenated?"
- **Ideal Follow-Up Answer:** concatenation lets a relationship emerge by chance, if at all; cross-attention architecturally forces the specific question Neupert physics actually asks.

Slide 8 — Technologies

- **Slide Purpose:** concrete, checkable credibility.
- **Psychological Goal:** "they could ship this."
- **Scientific Goal:** none — this is an engineering-trust slide.
- **Technical Goal:** resolved framework consistency; real, accurate tool names.
- **Key Message:** production-appropriate, ISRO-compatible.
- **Hidden Message:** nothing here should make you doubt the rest of the deck.
- **Visual Hierarchy:** flat — a logo row, deliberately with no single dominant element.
- **Information Hierarchy:** grouped by pipeline stage, matching Slides 5 and 7's groupings.
- **Common Mistakes:** naming a framework that contradicts what Slide 7's diagram already implied.
- **What Must Be Removed:** any tool listed that isn't actually used.
- **What Must Be Highlighted:** nothing — the goal is zero red flags, not a wow moment.
- **What Judges Will Ask:** "why this framework over the obvious alternative?"
- **Ideal Follow-Up Answer:** a short, confident, practical reason — team familiarity or deployment target — delivered the same way every time it's asked.

Slide 9 — Estimated Implementation Cost (recommended, despite optional)

- **Slide Purpose:** frugality and feasibility signal.
- **Psychological Goal:** "of course it's cheap — very ISRO."
- **Scientific Goal:** none.
- **Technical Goal:** single-GPU training window, real-time-capable inference, open-source licensing.
- **Key Message:** low-cost, high-rigor.

- **Hidden Message:** we understand ISRO has to operate this within real constraints, not just demo it once.
- **Visual Hierarchy:** relative cost comparison → line-item breakdown.
- **Information Hierarchy:** training cost, inference cost, licensing cost, in that order.
- **Common Mistakes:** inventing precise absolute rupee figures with false confidence; a padded cloud-infra line that contradicts the "lightweight" story told everywhere else.
- **What Must Be Removed:** any number the team can't defend if asked where it came from.
- **What Must Be Highlighted:** the contrast with a typical heavier ML pipeline, framed relatively, not as fabricated absolutes.
- **What Judges Will Ask:** "what would it cost to run this continuously, operationally?"
- **Ideal Follow-Up Answer:** the CUSUM-gated sentinel mode explicitly answers this — the system runs cheap and idle most of the time, escalating to full inference only when triggered.

Slide 10 — Thank You

- **Slide Purpose:** clean close.
- **Psychological Goal:** leave on a confident, non-needy note.
- **What Judges Will Ask:** the first open-floor question is almost always either the cross-attention-vs-Transformer objection (Slide 7) or the data-scarcity objection (Section 9 below) — be ready the moment this slide appears, not after.

SECTION 4 — THE NATIONAL IMPACT LAYER

Structural note first, because it matters more than the narrative itself: the official 10-slide template has **no dedicated National Impact slide**, and there shouldn't be an unofficial 11th one added — breaking template structure is its own selection risk. This narrative must be **woven into existing real estate**: a single escalating line closing out Slide 3 (USP), and the roadmap mention already planned for Slide 9. Anywhere else it appears, cut it.

Solar Physics → Space Weather → Infrastructure Protection → National Capability, the chain in one breath: a hard X-ray burst on the Sun becomes a soft X-ray glow seconds later (Neupert); that energy transfer, forecast early enough, becomes lead time; lead time becomes the difference between a satellite operator bracing for an anomaly and being blindsided by one; multiplied across enough satellites, telecom links, and GPS-dependent systems, that lead time becomes a measurable piece of national infrastructure resilience.

- **Short-term (hackathon window):** a working, explainable forecasting core on available Aditya-L1 and GOES data — proof that ISRO's own twin X-ray instruments can produce calibrated, physically-grounded forecasts without depending on external magnetogram products.
- **Mid-term (6–18 months, post-hackathon):** an operational pilot feeding existing space-weather advisory channels; expanding training data as Aditya-L1's own flare history accumulates; NDI used by solar physicists as a candidate-flagging tool for anomalous events worth a closer look.
- **Long-term (2–5 years):** extension toward the roadmap-only multi-payload platform (SWIS-ASPEX for CME/particle risk), feeding a broader Indian space-weather advisory capability serving satellite operators, telecom and GPS infrastructure, and polar-route aviation planning.
- **10-Year Vision:** an indigenous, instrument-native space-weather intelligence layer that doesn't depend on foreign data products for primary operational forecasting, scaling with every future Aditya-L1-class mission's payload suite — eventually supporting radiation-risk-aware planning for crewed missions.

Every stage above this hackathon's actual scope should be voiced as vision, explicitly, never as a current claim — the moment "10-year vision" starts sounding like "what we built," the section stops building trust and starts spending it.

SECTION 5 — THE VISUAL MASTERPLAN

Slide	Best Format	Why
1. Title	Hero Graphic	Template-fixed; first-impression real estate needs no added density.
2. Team	Icon Grid	Scannable, avoids text-heavy bios.
3. Opportunity/USP	Comparison Graphic	The slide's entire job is "different from" — a two-path contrast visualizes that directly.
4. Features	Icon Grid	Parallel structure for a capability inventory.
5. Process Flow	Data Flow Diagram	Literally a flow; the textbook-correct format.

Slide	Best Format	Why
6. Wireframe	Dashboard Mockup	Not a diagram — a designed interface; this is the one slide where this format is correct.
7. Architecture	Architecture Diagram	Custom, layered, hand-designed — never auto-generated-looking.
8. Technologies	Icon Grid	Flat, fast-scan, no hierarchy needed.
9. Cost	Comparison Graphic / simple Infographic	Relative framing benefits from contrast over a dense table.
10. Thank You	Hero Graphic	Template-fixed.

National-impact content (Section 4), where it appears on Slides 3 and 9, is text and tone, not a new visual element — resist the urge to add a Timeline or Roadmap graphic for it; that would be the 11th-slide mistake in graphic form.

SECTION 6 — MERMAID STRATEGY

Unchanged from the Story Bible, restated for this document's completeness: **Mermaid never appears in the judge-facing deck.** Its dense, planning-tool aesthetic rewards slow reading and signals "not designed for you" — a real, if subtle, credibility cost.

- **Where it belongs:** internal engineering docs (continue as-is), a hidden Q&A/appendix slide for "show me the full architecture" requests, and future notebook/README documentation.
- **Slides that deserve custom graphics instead:** Slide 5 (process flow) and Slide 7 (architecture) — both rebuilt as clean, hand-designed diagrams visually consistent with each other, six elements maximum, color-coded by function rather than by "what's new."

SECTION 7 — THE 8 MOST IMPORTANT MESSAGES

If judges remember only eight things, ranked:

1. Cross-attention here *is* the Neupert causal-energy-transfer relationship, computationally — not a metaphor for it.

2. It directly answers Challenge 15's literal ask: combined Hard+Soft X-ray forecasting from Aditya-L1's own instruments.
3. NDI can flag flares that violate expected physics — a discovery tool, not just a classifier.
4. CTE turns "trust me" into "watch the energy move" — second-by-second visual explainability.
5. Multi-horizon forecasts with calibrated uncertainty serve real operational decisions, not just a leaderboard metric.
6. The team shows genuine satellite-data literacy — GTI handling, fill-value auditing — that most entrants won't.
7. The system is low-cost and lightweight by design — frugal engineering, ISRO's own ethos.
8. The roadmap extends naturally to a national space-weather capability, without diluting the current, sharply-scoped build.

SECTION 8 — THE 3 KILLER SLIDES

Slide 3 (Opportunity/USP), Slide 7 (Architecture), and Slide 6 (Wireframe). Not Title, not Team, not Cost — selection gets decided in these three.

- **Slide 3** carries the core insight. Success looks like a judge being able to repeat the one-liner back without rereading it.
- **Slide 7** carries technical credibility. Success looks like a domain judge nodding instead of squinting — the test for whether the six-macro-block compression worked.
- **Slide 6** carries memorability. Success looks like a judge saying "can I see this running" out loud, unprompted.

Effort allocation should be disproportionate: these three earn the design hours, the rehearsal time, and the most scrutiny before submission. Slides 1, 2, 4, 8, 9, and 10 need to be *clean and correct* — that's the full bar for them. Spending equal polish across all ten slides is a worse use of six days than spending most of it on three.

SECTION 9 — THE RED TEAM ATTACK

Acting as a hostile review board, not a supportive one.

#	Objection	Best Response
1	Attention-based flare forecasting already exists — what's actually new?	Concede the broader landscape; the narrow, defensible claim is cross-modal Hard↔Soft attention bound to the Neupert relationship using Aditya-L1's own instruments — not "attention" itself.
2	NDI is derived from the model's own training loss — isn't "deviation" just where the model fits poorly, not new physics?	Honest concession: NDI alone can't distinguish model error from real anomaly. The claim is narrower — NDI is a candidate-flagging tool for human follow-up, validated by checking flagged events against known unusual flares, not an autonomous discovery claim.
3	The real build window is ~2.5–3 weeks post-shortlist, not 2+ months — is this buildable?	Scope the deliverable to match the calendar explicitly: core Notebook 2 plus a subset of diagnostics by the finale, with the rest labeled "designed, not yet built."
4	Does Aditya-L1 actually have enough M/X-class flares to train on?	State this as the single biggest scientific risk, mitigated by GOES pre-training, and report the actual flare count transparently rather than implying abundance.
5	Is the GOES→Aditya-L1 cross-calibration validated, or assumed?	Concede it's currently a hypothesis; name the validation step (overlap-window comparison) as a defined next step, not a solved problem.
6	MC Dropout is a known weak approximation to real Bayesian uncertainty — why use it?	Acknowledge the limitation; frame it as a practical first pass appropriate for the timeline, with proper calibration (e.g. conformal methods) as the named next upgrade.
7	Are the NDI risk cutoffs (0.3/0.6/0.8) physically motivated or arbitrary?	Be honest if they're currently heuristic; commit to calibrating them against historical NOAA event statistics rather than presenting them as physically derived if they aren't.
8	How is this different from just concatenating Hard and Soft X-ray into any Transformer?	Concatenation lets a relationship emerge by chance, if at all; cross-attention architecturally forces the specific question Neupert physics asks.
9	This is a 16-component system — how much is actually built versus planned?	Be explicit and unembarrassed: Notebook 1 is frozen; everything else is designed-but-unbuilt. False precision about completion is a bigger risk than honest incompleteness.

#	Objection	Best Response
10	Is the dashboard mockup real data or illustrative?	State plainly which it is; if illustrative, point to what's already validated in Notebook 1 as the real evidence base.
11	NOAA's catalog has known completeness and timing-precision limits — how does that affect ground truth?	Acknowledge the catalog isn't perfect; note label noise is a shared limitation across the entire field, not unique to this project.
12	Why this ML framework over the obvious alternative?	A short, practical, non-defensive reason — team familiarity or deployment target — and the same answer every time it's asked.
13	What happens to this after the hackathon ends — who maintains it?	Have a real, even modest answer ready: continued development, intent to publish, or handoff documentation — not silence.
14	Has this been benchmarked against any baseline (e.g. simple logistic regression on the same features)?	If not yet done, commit to it as a near-term step — one honest baseline number outweighs five unvalidated claims.
15	Why these specific five features and not others?	Each ties to a named diagnostic already used in solar-physics literature (Neupert; Dennis & Zarro) — state the lineage rather than presenting the feature set as arbitrary.
16	Is the team's experience sufficient to execute this in the real timeline?	Lean on what's concretely already done — the frozen Notebook 1 — as evidence of execution capacity, rather than promises about the rest.
17	Could this architecture work just as well on GOES data alone — how Aditya-L1-specific is it, really?	Be honest the architecture is data-agnostic in principle; what's Aditya-L1-specific is the literal challenge fit and the GTI/instrument-format handling — that's the relevance strength, not an architectural necessity.
18	Does accuracy degrade sharply at the 24h horizon, and is that being hidden?	Commit to reporting horizon-specific metrics transparently, even where longer-horizon performance is weaker.

#	Objection	Best Response
19	Is "physics-informed" a marketing label, or does removing the Neupert loss term measurably hurt performance?	Commit to running the actual ablation (with vs. without the physics loss) and reporting it honestly — this single piece of evidence would be more convincing than everything else in the deck combined, and it doesn't exist yet.
20	With hundreds of submissions, why should this one survive a 60-second skim?	This is the meta-objection the rest of this document exists to answer — the response is the discipline of Sections 1–8, not a new argument invented here.

SECTION 10 — THE FINAL SELECTION AUDIT

Dimension	Score
Innovation	76/100
Scientific Merit	83/100
ISRO Relevance	95/100
Presentation Potential	85/100 *(conditional — reflects the upside if Sections 1–8's discipline is followed; no slides exist yet)*
National Impact	80/100 *(strong narrative connection; several of the eight impact domains are roadmap, not demonstrated — score reflects narrative strength, not current scope)*

Selection Probability: ~75% to advance past the idea round if this document's curation discipline is followed end to end; drops to 45–55% if the underlying complexity gets presented unfiltered.

Most Likely Rejection Reason: information overload obscuring the core insight during a fast skim, compounded by any surviving overclaim language ("first," "novel," "unprecedented") that a domain-literate reviewer catches and discounts everything after.

Most Likely Selection Reason: precise, verifiable architectural alignment with Challenge 15's literal wording, paired with genuine, checkable physics grounding — a combination rare enough among likely competing submissions to stand out on its own, without needing exaggeration.

Final Recommendations:

1. If any time exists before the finale, run the physics-loss ablation (objection #19) — it's the single most convincing piece of evidence the project could produce and currently doesn't exist.
2. Resolve the framework inconsistency (PyTorch idiom vs. Keras export) everywhere, before anything ships.
3. Commit to one primary frame — forecasting — with NDI as the supporting beat, not a co-equal headline.
4. Strike every unscoped superlative; replace with the narrower, defensible novelty claim from Section 9, objection #1.
5. Weave national impact into Slides 3 and 9 only — never add an eleventh slide for it.
6. Protect Slides 3, 6, and 7 with disproportionate design and rehearsal time; the other seven only need to be clean.

This document is the direct source for slide creation. If a future draft slide contradicts a decision made here, the draft is wrong — return to this file, resolve the conflict, then regenerate.